

Helium

He (He I). Strongest classified transitions below 2 μm (NIST): 53 levels, 79 lines.

He⁺ ionisation limit 198310.67 cm⁻¹ = 24.58739 eV (50.426 nm)

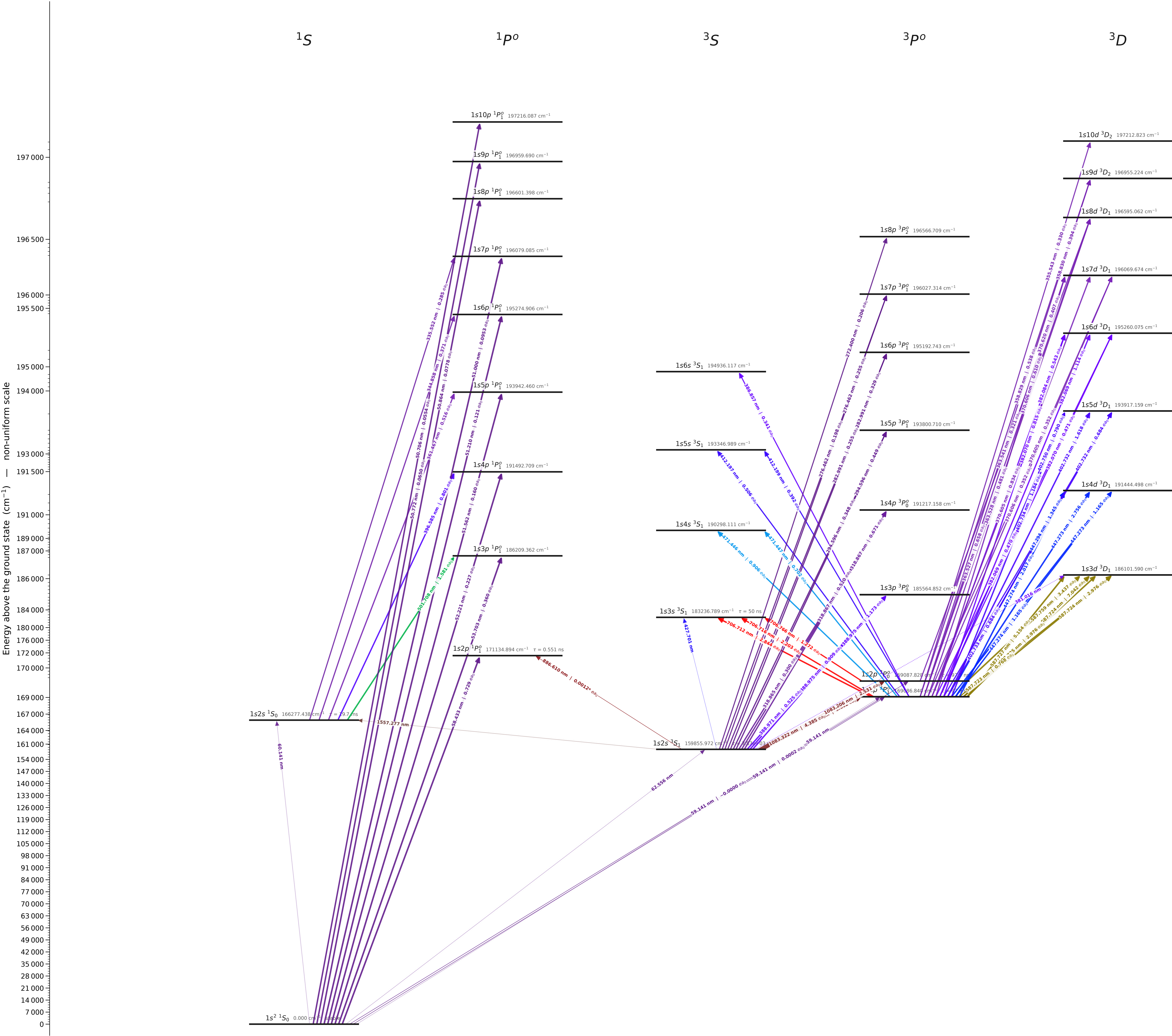
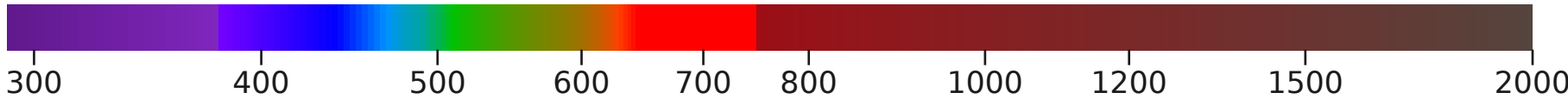
Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J || e r || J' \rangle$ in ea₀

Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.

Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.

Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle J e r J' \rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
1s ² 1S ₀ - 1s2p 3P ₀ ^o	59.1409	-	5069.1256	0.000	1.000e-02	1.626	0	strictly forbidden decay of 2 3P0
1s ² 1S ₀ - 1s2p 3P ₁ ^o	59.1412	-	5069.0959	0.000	1.770e+02	1.626	0	59.1 nm intercombination decay of 2 3P1
1s ² 1S ₀ - 1s2p 3P ₂ ^o	59.1412	-	5069.0937	-	3.240e-01	1.626	0	M2 decay of 2 3P2 to the ground state
1s ² 1S ₀ - 1s2s 1S ₀	60.1405	-	4984.8722	-	5.076e+01	8.079e-06	100	two-photon
1s ² 1S ₀ - 1s2s 3S ₁	62.5563	-	4792.3615	-	1.271e-04	2.022e-11	100	62.6 nm M1 decay of the metastable 2 3S1 state
1s2s 3S ₁ - 1s3d 3D ₁	381.0160	380.9078	786.823850	-	-	-	-	762 nm two-photon line
1s2s 3S ₁ - 1s3p 3P ₀ ^o	388.9707	388.8605	770.732839	0.525	9.475e+06	-	-	389 nm line
1s2s 3S ₁ - 1s3s 3S ₁	427.7010	427.5807	700.939271(5)	-	7.000e-09	3.183	0	427.7 nm forbidden
1s2p 3P ₀ ^o - 1s3d 3D ₁	587.7595	587.5966	510.059755	3.437	3.928e+07	-	-	587.6 nm
1s2p 3P ₁ ^o - 1s3s 3S ₁	706.7163	706.5215	424.204793(5)	2.203	9.284e+06	3.183	46.4	706.7 nm line
1s2s 3S ₁ - 1s2p 1P ₁ ^o	886.6095	886.3661	338.133594	0.001*	1.442e+00	288.8	0	887 nm spin-forbidden line
1s2s 3S ₁ - 1s2p 3P ₀ ^o	1083.2058	1082.9091	276.764095	2.531	1.022e+07	1.626	100	1083 nm D0 line
1s2s 3S ₁ - 1s2p 3P ₁ ^o	1083.3217	1083.0250	276.734478	4.385	1.022e+07	1.626	100	1083 nm D1 line
1s2s 3S ₁ - 1s2p 3P ₂ ^o	1083.3306	1083.0340	276.732187	5.662	1.022e+07	1.626	100	1083 nm laser-cooling / MOT / imaging line
1s2s 3S ₁ - 1s2s 1S ₀	1557.2768	1556.8514	192.510702	-	9.100e-08	8.079e-06	0	1557 nm doubly-forbidden ortho-para line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

³He hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
1s2s 3S ₁	-4493.134118(11)	-	F=1/2: +4493.13 F=3/2: -2246.57

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
1s2s 3S ₁ - 1s2p 3P ₀ ^o	1083.206	3-4	33668.074(5)
1s2s 3S ₁ - 1s2p 3P ₂ ^o	1083.331	4-3	33668.445(3)
1s2s 3S ₁ - 1s2s 1S ₀	1557.277	4-3	8034.3672(23)

Shift = ν(first isotope) - ν(second isotope).